

MIAE 215 · Mini-Course Lesson 1

Software Installation & Development Environment Setup

Concordia University · Department of Mechanical, Industrial & Aerospace Engineering (MIAE)

Source: [Lesson 1 Web Page](#)

Table of Contents

- 1. [Required Software Package Checklist](#)
- 2. [The Code::Blocks IDE & MinGW Compiler Toolchain](#)
- 3. [Text Editors & Archive Utilities](#)
- 4. [Verification: Building Your First Test Executable](#)

1. Required Software Package Checklist

Before writing C++ programs, engineers require five foundational software utilities:

Tool	Recommended Software	Function in MIAE 215
C++ IDE & Compiler	Code::Blocks 17.12 / 20.03 (with MinGW)	Integrated development environment, syntax highlighting, compiler & debugger.
Video Player	VLC Media Player (videolan.org)	Plays high-bitrate <code>.mp4</code> lecture screencasts with full audio codec support.
PDF Reader	Adobe Acrobat Reader (get.adobe.com/reader)	Renders comprehensive topic guides, slides, and syllabus tables without font glitches.
Programmer's Editor	Notepad++ (notepad-plus-plus.org)	Lightweight source code inspection, line numbering, and script editing.
Archive Utility	7-Zip (7-zip.org) or WinRAR	Extracts <code>.rar</code> and <code>.zip</code> weekly course packs and software bundles.

2. The Code::Blocks IDE & MinGW Compiler Toolchain

The primary recommended development platform for MIAE 215 is **Code::Blocks** paired with the **MinGW GNU GCC/G++** compiler.





Installation Steps on Windows

1. Download `codeblocks-17.12mingw-setup.exe` (or `codeblocks-20.03mingw-setup.exe`).
2. **Crucial:** Ensure you download the installer that includes `mingw` in the filename! A setup without MinGW includes only the editor without the actual C++ compiler.
3. Run the installer and accept default settings (installing to `C:\Program Files\CodeBlocks`).
4. **Portable Alternative:** If you cannot install software on your machine, a standalone pre-configured portable version is provided in this repository at: `05 - Software & Flowcharts/CodeBlocks_17.12_portable/`

3. Text Editors & Archive Utilities

- **Notepad++:** Prof. Gordon demonstrates Notepad++ for quickly reading `.cpp` files without launching a heavy IDE. Download demo video: http://users.encs.concordia.ca/~bwgordon/notepad++_demo.mp4.
- **7-Zip / WinRAR:** All course code packs from the professor are distributed as `.rar` archives. Extract files using right-click \$ o\$ "Extract Here".

4. Verification: Building Your First Test Executable

To verify that your compiler is operating correctly:

1. Open Code::Blocks \$ o\$ **File \$ o\$ New \$ o\$ Project...**
2. Select **Console Application** \$ o\$ choose **C++**.
3. Set project title as `test_installation` and save in a clean directory on your local drive.
4. Open `main.cpp` and press **F9** (Build and Run).
5. If a black console window appears displaying `"Hello world!"`, your toolchain is 100% operational!